

16	When coherent monochromatic light falls on double slits, an interference pattern is observed on a screen some distance from the slits. The fringe separation can be increased by	
	A	Decreasing the distance between the screen and the slits.
	B	Increasing the distance between the slits.
	C	Using monochromatic light of lower frequency.
	D	Immersing the whole set up in water.

	Ans: C, $x = \frac{\lambda D}{a}$, hence decreasing D, increasing a causes x to be lower. Lower frequency means higher λ resulting in higher x. Immersing set up in water causes speed of light to decrease which in turn causes wavelength to decrease (frequency stays the same) resulting in lower x.
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